# CHEM 103 Unit 2 Summary: Hybridization and Molecular Orbital Theory

Focused review notes | Fall 2026

**Based mainly on CHEM103-U2P4-Bonding(DEFG) (lecture pages 37-61) and CHEM103-U2P5-Bonding(FGH) (lecture pages 58-62).**

## 中文速览

- 判断杂化：先画 Lewis 结构，再数目标原子的电子组。2、3、4 个电子组分别对应 sp、$sp^2$、$sp^3$；多重键整体只算一个电子组。
- 单键 = 1σ；双键 = 1σ + 1π；三键 = 1σ + 2π。σ 键来自沿键轴的端对端重叠，π 键来自平行未杂化 p 轨道的侧向重叠。
- MO 图中，成键轨道降低能量，反键轨道升高能量并以 * 标记。
- 键级 = ½(成键电子数 - 反键电子数)。键级越大，通常键越强、越短。
- 有未成对电子为顺磁性（paramagnetic）；全部成对为抗磁性（diamagnetic）。$O_2$ 的两个未成对电子位于 π*2p 轨道。
- $B_2$、$C_2$、$N_2$ 的 2p 能级顺序中 π2p 低于 σ2p；$O_2$、$F_2$、$Ne_2$ 中 σ2p 低于 π2p。

## Learning goals

After reviewing this note, you should be able to:

1. determine the hybridization of an atom from its electron-group geometry;
2. describe single, double, and triple bonds using sigma (σ) and pi (π) overlap;
3. construct and fill MO diagrams for homonuclear diatomic molecules and ions;
4. calculate bond order and predict relative stability, bond strength, bond length, and magnetic behavior;
5. explain the difference between localized valence-bond descriptions and delocalized molecular-orbital descriptions.

## 1. Valence Bond (VB) Theory

### 1.1 Basic idea

A covalent bond forms when atomic orbitals on two atoms overlap and two electrons with opposite spins become localized in the region between the nuclei.

- Greater constructive (in-phase) overlap gives a stronger bonding interaction.
- The orientation and shape of the orbitals control how effectively they overlap.
- Ordinary isolated-atom s and p orbitals do not directly explain all observed

molecular geometries. Hybrid orbitals provide a useful bonding model.

### 1.2 Meaning of hybridization

Hybridization is the mathematical mixing of atomic orbitals on the **same atom** to produce a new set of equivalent hybrid orbitals with useful orientations.

Important points:

- the number of hybrid orbitals formed equals the number of atomic orbitals mixed;
- hybrid orbitals form σ bonds or hold lone pairs;
- any p orbitals not included in the mixing remain unhybridized and can form π bonds;
- hybridization is a model used to explain observed geometry, not a literal physical

event that happens to an isolated atom before bonding.

## 2. Predicting Hybridization

### 2.1 Fast method

1. Draw the best Lewis structure, including resonance when relevant.
2. Choose the atom whose hybridization is required.
3. Count its electron groups (steric number):

- each single, double, or triple bond counts as **one** electron group;
- each lone pair counts as **one** electron group.

4. Match the number of electron groups to the electron-group geometry and

hybridization.

| Electron groups | Electron-group geometry | Ideal angles | Orbitals mixed | Hybridization | Unhybridized p orbitals |
|---|---|---|---|---|---|
| 2 | linear | 180° | one s + one p | sp | 2 |
| 3 | trigonal planar | 120° | one s + two p | $sp^2$ | 1 |
| 4 | tetrahedral | 109.5° | one s + three p | $sp^3$ | 0 |
| 5 | trigonal bipyramidal | 90°, 120°, 180° | one s + three p + one d | $sp^3d$ | 0* |
| 6 | octahedral | 90°, 180° | one s + three p + two d | $sp^3d^2$ | 0* |

*In the introductory hybridization model used for molecules such as $PCl_5$ and $SF_6$. Second-period elements have no accessible 2d orbitals and cannot expand their octets.

### 2.2 Electron-group geometry versus molecular shape

Hybridization follows **electron-group geometry**, which includes lone pairs. Molecular shape describes only the positions of atoms.

- $CH_4$: four groups → tetrahedral electron geometry, tetrahedral shape, C is $sp^3$.

- $NH_3$: four groups → tetrahedral electron geometry, trigonal-pyramidal shape, N is $sp^3$; one $sp^3$ orbital holds the lone pair.
- $H_2O$: four groups → tetrahedral electron geometry, bent shape, O is $sp^3$; two $sp^3$ orbitals hold lone pairs.

### 2.3 Core examples from the lecture

| Molecule | Central atom | Electron groups | Hybridization | Geometry / shape | Bonding picture |
|---|---|---|---|---|---|
| $BeCl_2$ | Be | 2 | sp | linear | two Be sp orbitals overlap with Cl orbitals |
| $BF_3$ | B | 3 | $sp^2$ | trigonal planar | three B $sp^2$ orbitals form three σ bonds; one p remains |
| $CH_4$ | C | 4 | $sp^3$ | tetrahedral | four equivalent C($sp^3$)-H(1s) σ bonds |
| $NH_3$ | N | 4 | $sp^3$ | trigonal pyramidal | three σ bonds plus one lone pair in $sp^3$ orbitals |
| $H_2O$ | O | 4 | $sp^3$ | bent | two σ bonds plus two lone pairs in $sp^3$ orbitals |
| $PCl_5$ | P | 5 | $sp^3d$ | trigonal bipyramidal | five σ bonds in the introductory model |
| $SF_6$ | S | 6 | $sp^3d^2$ | octahedral | six σ bonds in the introductory model |

## 3. Sigma and Pi Bonds

### 3.1 Sigma (σ) bonds

A σ bond results from end-to-end overlap along the internuclear axis.

- Electron density is cylindrically symmetric around the bond axis.
- Every single bond is one σ bond.
- The first bond in a double or triple bond is also a σ bond.
- A σ bond usually permits rotation if no other structural restriction is present.

Possible overlaps include s-s, s-p, p-p, s-hybrid, p-hybrid, and hybrid-hybrid.

### 3.2 Pi (π) bonds

A π bond results from side-by-side overlap of two parallel, unhybridized p orbitals.

- Electron density lies in two regions on opposite sides of the bond axis.
- A π bond restricts rotation because the two p orbitals must remain parallel.
- π overlap is generally weaker than σ overlap because it is less direct.

### 3.3 Bond-count rules

| Bond type | σ bonds | π bonds |
|---|---|---|
| single | 1 | 0 |
| double | 1 | 1 |
| triple | 1 | 2 |

## 3.4 Carbon examples

### Ethane, $C_2H_6$

- Each C is $sp^3$.
- The C-C bond is one $sp^3$-$sp^3$ σ bond.
- The six C-H bonds are $sp^3$-1s σ bonds.
- Total: 7 σ and 0 π bonds.

### Ethene, $C_2H_4$

- Each C is $sp^2$ and approximately trigonal planar.
- One $sp^2$ orbital from each C forms the C-C σ bond.
- The unhybridized p orbital on each C overlaps sideways to form one π bond.
- Total: 5 σ and 1 π bond.
- Rotation about C=C would break the π overlap, so rotation is restricted.

### Ethyne (acetylene), $C_2H_2$

- Each C is sp and linear.
- One sp orbital from each C forms the C-C σ bond.
- Two pairs of mutually perpendicular unhybridized p orbitals form two π bonds.
- Total: 3 σ and 2 π bonds.

### Acetic acid, $CH_3COOH$

Using the localized Lewis/VB description expected at this level:

- methyl C: $sp^3$;
- carbonyl C: $sp^2$;
- carbonyl O: $sp^2$;
- hydroxyl O: $sp^3$ by direct electron-group counting;
- C=O contains one σ and one π bond; all other drawn single bonds are σ bonds.

The carboxyl group also has resonance/delocalization. A more advanced description treats the hydroxyl O lone pair as conjugated with the carbonyl π system.

# 4. Molecular Orbital (MO) Theory

## 4.1 VB theory compared with MO theory

| Valence Bond theory | Molecular Orbital theory |
|---|---|
| treats a bond as localized between two atoms | allows electrons to be delocalized over the molecule |
| uses overlap of atomic or hybrid orbitals | combines atomic orbitals to form molecular orbitals |
| describes σ and π bonds | describes bonding and antibonding MOs |
| is especially useful for molecular shape and local bonding | is especially useful for bond order and magnetism |
| often uses resonance structures for delocalization | represents delocalization directly in the MOs |

## 4.2 Combining atomic orbitals

When two compatible atomic orbitals combine, they produce two molecular orbitals:

1. **Bonding MO:** constructive combination; electron density increases between

the nuclei; lower energy.

2. **Antibonding MO:** destructive combination; a node occurs between the nuclei;

higher energy; marked with an asterisk (*).

Therefore:

**number of AOs combined = number of MOs produced**

Examples:

- 1s + 1s → σ1s and σ*1s
- end-to-end 2p + 2p → σ2p and σ*2p
- side-by-side 2p + 2p → π2p and π*2p

The two π orbitals formed from the two perpendicular p directions have the same energy; they are **degenerate**.

## 4.3 Filling an MO diagram

Use the same electron-filling rules as for atomic orbitals:

- **Aufbau principle:** fill lower-energy MOs first;
- **Pauli exclusion principle:** at most two electrons per orbital, with opposite spins;
- **Hund's rule:** fill degenerate orbitals singly with parallel spins before pairing.

For the homonuclear second-period species emphasized in this course, count the valence electrons from both atoms and adjust for charge. The 1s core orbitals are often omitted because their bonding and antibonding contributions cancel.

# 5. MO Energy Ordering for Second-Period Diatomics

The order of the 2p MOs changes because of 2s-2p mixing.

## $B_2$, $C_2$, and $N_2$: with significant 2s-2p mixing

From low to high energy:

**σ2s < σ*2s < π2p < σ2p < π*2p < σ*2p**

Here, π2p is below σ2p.

### $O_2$, $F_2$, and $Ne_2$: without significant 2s-2p mixing

From low to high energy:

**σ2s < σ*2s < σ2p < π2p < π*2p < σ*2p**

Here, σ2p is below π2p.

### Memory aid

- **B-C-N:** π comes before σ in the 2p bonding levels.
- **O-F-Ne:** σ comes before π in the 2p bonding levels.
- The change occurs between $N_2$ and $O_2$.

## 6. Bond Order, Stability, and Magnetism

### 6.1 Bond order

**Bond order = 1/2 × (number of bonding electrons - number of antibonding electrons)**

- Bond order > 0: the species has net bonding and is stable relative to separated atoms.
- Bond order = 0: no net bond is predicted.
- Higher bond order generally means a stronger, shorter bond and a larger bond energy.
- Electrons in nonbonding orbitals, when present, do not change bond order.

### 6.2 Magnetic behavior

- one or more unpaired electrons → **paramagnetic**;
- all electrons paired → **diamagnetic**.

Do not infer magnetism from bond order alone. Inspect the electron occupancy of the individual MOs.

### 6.3 Key homonuclear diatomic results

| Species | Valence electrons | Bond order | Magnetism | Main conclusion |
|---|---|---|---|---|
| $H_2$ | 2 | 1 | diamagnetic | stable single bond |
| $He_2$ | 4 | 0 | diamagnetic | no net bond; not stable in the basic model |
| $Li_2$ | 2 in 2s levels | 1 | diamagnetic | stable |
| $Be_2$ | 4 in 2s levels | 0 | diamagnetic | no net bond in the basic model |
| $B_2$ | 6 | 1 | paramagnetic | two unpaired electrons in π2p |
| $C_2$ | 8 | 2 | diamagnetic | all occupied MOs paired |
| $N_2$ | 10 | 3 | diamagnetic | very strong, short bond |
| $O_2$ | 12 | 2 | paramagnetic | two unpaired electrons in π*2p |
| $F_2$ | 14 | 1 | diamagnetic | weaker and longer than $O_2$ and $N_2$ |
| $Ne_2$ | 16 | 0 | diamagnetic | no net bond |

The paramagnetism of $O_2$ is a major success of MO theory and is not predicted correctly by a simple localized Lewis structure.

# 7. Diatomic Ions

## 7.1 General procedure

1. Draw the correct neutral-species MO ordering.
2. Fill the neutral species.
3. For a cation, remove electrons from the highest occupied MO.
4. For an anion, add electrons to the lowest unoccupied MO.
5. Recalculate bond order and count unpaired electrons.

The effect of ionization depends on the type of orbital changed:

| Electron change | Effect on bond order |
|---|---|
| add one electron to a bonding MO | +0.5 |
| remove one electron from a bonding MO | -0.5 |
| add one electron to an antibonding MO | -0.5 |
| remove one electron from an antibonding MO | +0.5 |

## 7.2 Useful examples

| Species | Change from neutral molecule | Bond order | Magnetic behavior |
|---|---|---|---|
| $N_2^+$ | remove one electron from σ2p | 2.5 | paramagnetic |
| $N_2^-$ | add one electron to π*2p | 2.5 | paramagnetic |
| $O_2^+$ | remove one electron from π*2p | 2.5 | paramagnetic |
| $O_2^-$ | add one electron to π*2p | 1.5 | paramagnetic |
| $O_2^{2-}$ | add two electrons to π*2p | 1 | diamagnetic |

For the oxygen series, increasing occupation of π*2p weakens and lengthens the O-O bond:

**$O_2^+ > O_2 > O_2^- > O_2^{2-}$ in bond order and bond strength**

The bond-length order is the reverse.

# 8. Ground States, Excited States, and Delocalization

## 8.1 Excited molecular states

The normal Aufbau filling is the molecular ground state. Absorption of energy can promote an electron to a higher-energy MO.

For $H_2$:

- ground state: $(\sigma 1s)^2$, bond order = 1;
- simple excited configuration: $(\sigma 1s)^1(\sigma^* 1s)^1$, bond order = 0.

Promotion into an antibonding MO weakens the bond.

## 8.2 Delocalized π bonding

VB theory usually represents delocalization using several resonance contributors. The real molecule does not switch back and forth between them; its structure is a resonance hybrid.

MO theory treats the relevant electrons as occupying orbitals extended across several atoms. This directly describes the delocalized π systems in molecules such as ozone ($O_3$) and benzene ($C_6H_6$).

# 9. Exam Workflow

## Hybridization / bonding-description question

1. Draw the Lewis structure and all important resonance contributors.
2. Count electron groups around each requested atom.
3. Assign electron geometry and hybridization.
4. Identify the orbitals that form each σ bond.
5. For each multiple bond, identify the unhybridized p orbitals that form the π bond(s).
6. State geometry, approximate angles, and any restriction on rotation.

## MO-diagram question

1. Count total valence electrons and adjust for charge.
2. Choose the correct MO ordering: B-C-N or O-F-Ne.
3. Draw the atomic-orbital sides and central MO levels.
4. Fill electrons using Aufbau, Pauli, and Hund.
5. Write the MO electron configuration if requested.
6. Calculate bond order.
7. Count unpaired electrons and state paramagnetic or diamagnetic.
8. Compare strength and length only after determining bond order.

# 10. Common Mistakes

- Counting a double or triple bond as more than one electron group for hybridization.
- Assigning hybridization from molecular shape while ignoring lone pairs.
- Saying a double bond contains two σ bonds; it contains one σ and one π.
- Saying a triple bond contains three σ bonds; it contains one σ and two π.
- Using hybrid orbitals to form π bonds; π bonds require unhybridized p orbitals.
- Forgetting that π2p and σ2p reverse order between $N_2$ and $O_2$.
- Pairing electrons in degenerate π orbitals before applying Hund's rule.
- Counting total electrons in the bond-order formula without separating bonding and antibonding electrons.
- Assuming a positive bond order automatically means diamagnetic.
- Removing an electron from an atomic orbital rather than the highest occupied MO when forming a molecular ion.

# 11. One-Minute Review

- 2, 3, 4 electron groups → sp, $sp^2$, $sp^3$.
- single = 1σ; double = 1σ + 1π; triple = 1σ + 2π.
- σ overlap is end-to-end; π overlap is side-by-side.
- bonding MOs lower energy; antibonding MOs (*) raise energy and contain a node.
- bond order = ½(bonding electrons - antibonding electrons).
- unpaired electrons mean paramagnetic.
- $B_2$/$C_2$/$N_2$: π2p below σ2p.
- $O_2$/$F_2$/$Ne_2$: σ2p below π2p.
- $O_2$ is paramagnetic because it has two unpaired electrons in π*2p.
- higher bond order usually means a stronger and shorter bond.